A PROJECT REPORT

On

Development Of Electric ATV

Submitted By

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In fulfillment for the award of the Diploma of

Diploma of Engineering

In

MECHANICAL ENGINEERING

SCHOOL OF TECHNOLOGY & ENGINEERING (SOTE)

ITM (SLS) BARODA UNIVERSITY, VADODARA

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CERTIFICATE

This is to certify that the report submitted along with the project entitled **"Development of Electric All-Terrain Vehicle (EV-ATV)"** has been carried out by **Param Dave (23T17011), Dev Bhadauria (23T17001), Tirth Darji (23T17010), Pritesh Chavda (23T17009), Dhairya Bhosale (23T17005), Rayan Patel (23T17022), Bhaumik Bhaliya (23T17002), Prem Bhaliya (23T17004), Mayur Solanki (24T17511)** under my guidance in partial fulfillment for the **Diploma in Mechanical Engineering**, VIth Semester, School of Technology & Engineering (SOTE), ITM (SLS) Baroda University, during the academic year **2025–2026**. These students have successfully completed the project activity under my guidance.

SIGNATURE

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DECLARATION

We, Param Dave, Dev Bhadauria, Tirth Darji, Pritesh Chavda, Dhairya Bhosale, Rayan Patel, Bhaumik Bhaliya, Prem Bhaliya and Mayur Solanki, the students of Diploma in Mechanical Engineering, having enrolment numbers 23T17011, 23T17001, 23T17010, 23T17009, 23T17005, 23T17022, 23T17002, 23T17004 and 24T17511, enrolled at ITM (SLS) Baroda University, Vadodara, hereby certify and declare the following:

1. We have defined our project entitled "Development of Electric All-Terrain Vehicle (EV-ATV)" based on inputs at ITM (SLS) Baroda University and each of us will make significant efforts to solve the challenges. We will carry out the project work at our college or at any location under the direct and consistent monitoring of our guides. We will adopt all ethical practices to share credit amongst all contributors based on their contributions during the project work.
2. We have not purchased the solutions developed by any third party, and all efforts have been made by us under the guidance of our project guides.
3. The project work is original and has not been copied from any previously done projects.
4. ITM (SLS) Baroda University, to the best of our knowledge, is a genuine institution engaged in professional and academic activities.
5. We understand and accept that if the above declaration is found to be untrue, it may result in cancellation of our project work, including failure in the subject.

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We would like to express our sincere and heartfelt gratitude to all those who have contributed to the successful completion of our project entitled “Development of Electric All-Terrain Vehicle (EV-ATV)”.

First and foremost, we would like to express our deep sense of gratitude to our respected guides Mr. Om Patel, Asst. Prof. Nayan Kaneriya, and Dr. Mahesh Patel for their constant support, invaluable guidance, and continuous encouragement throughout the duration of this project. Their expert advice, constructive suggestions, and motivation played a crucial role in helping us overcome various challenges and successfully complete this project.

We would also like to extend our sincere thanks to all the teaching and non-teaching staff members of the Department of Mechanical Engineering, ITM (SLS) Baroda University, for their cooperation, assistance, and support at different stages of the project.

We are equally grateful to our institution for providing us with the necessary facilities, resources, and a conducive environment to carry out this project work effectively.

Lastly, we are deeply indebted to the college for providing us with the opportunity to work on this project and for fostering an environment that encourages learning, innovation, and technical development.

ABSTRACT

Due to the increasing need for eco-friendly and cost-effective transportation, electric vehicles are gaining importance in the automobile industry. This project deals with the development of an Electric All-Terrain Vehicle (EV ATV) using a scrap ATV chassis, without an internal combustion engine (ICE) and fuel system.

In this project, a scrap chassis is selected and modified to accommodate an electric drive system consisting of a battery pack, electric motor, motor controller, and associated electrical components. Since the ICE and fuel system are not used, the vehicle operates solely on electrical energy. Existing systems such as steering, braking, and suspension are retained and suitably modified to meet the requirements of electric operation.

The electric motor converts electrical energy into mechanical energy to drive the vehicle, resulting in smooth, silent, and emission-free operation. This project highlights the effective reuse of scrap components and demonstrates a practical and economical method for developing an electric ATV. The developed EV ATV is suitable for academic learning, offroad applications, and experimental purposes.

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1.INTRODUCTION

1.1 Overview

An Electric Vehicle (EV) is a vehicle that operates with mainly electrical energy that comes from stored rechargeable batteries rather than traditional fuels, including gasoline and diesel.

Electric energy stored within the vehicle's batteries is sent to the electric motor, where it is converted into a mechanical form of energy, which is used to move the vehicle's wheels.

An electric vehicle essentially consists of five main components: An electric motor(s), Battery Pack(s), Motor Controllers, a Battery Management System (BMS), and a Charging Station.

Because EVs do not use an Internal Combustion Engine (ICE), they do not produce any exhaust emissions while operating, thus reducing air pollution and dependency upon fossil fuels.

The large amount of torque produced; the simplicity of design; and the efficiency in delivering power through the electric motor, make electric vehicles a perfect fit for offroad applications, such as electric All-Terrain Vehicles (ATVs).

Figure 1.1 Electric ATV Overview

1.2 Introduction to ATV

The All-Terrain Vehicle (ATV) is an off-road vehicle designed to operate on uneven surfaces such as mud, sand, gravel, and rocky terrain. This type of vehicle is characterized by a compact and lightweight structure, wide low-pressure tires, high ground clearance, and a simple suspension system, enabling it to perform efficiently under challenging conditions.

The Electric ATV (EV-ATV) uses an electric motor powered by a rechargeable battery pack instead of a traditional internal combustion (gasoline) engine.
The electric motor produces high torque at low speeds, which is essential for off-road applications.

Electric ATVs offer several advantages over conventional gasoline ATVs, including zero tailpipe emissions, lower noise levels, higher energy efficiency, and reduced maintenance requirements.
Properly designed electric ATVs are well-suited for campus transportation and recreational applications.

1.3 Problem Identification

Conventional ATVs powered by internal combustion engines consume fossil fuels and produce harmful exhaust emissions, leading to air and noise pollution.

• High fuel cost, frequent maintenance, and complex engine components increase the operating cost of ICE-based ATVs.

• Many ATV chassis are discarded as scrap due to engine failure or outdated technology, resulting in wastage of usable materials.

• Reusing a scrap ATV chassis without an ICE and fuel system can reduce cost and material wastage.

• Hence, the problem identified is to develop an electric ATV by integrating an efficient electric drive system suitable for off-road applications.

1.4 Scope of Project

- Existing Condition and Aim of Redevelopment

Existing Condition

The existing vehicle was originally a fully functional ICE-based all-terrain vehicle. However, due to long-term abandonment and lack of maintenance, it is currently in a non-operational condition.

Inspection reveals rusted and corroded chassis joints, worn and misaligned steering, suspension, braking, and drivetrain components. Additionally, several engine and fuel system parts are either damaged or missing.

The electrical wiring is outdated and exposed, making the vehicle unsafe for operation. Therefore, the vehicle does not meet current safety, performance, and environmental standards.

Aim of Redevelopment

The aim of redevelopment is to convert the abandoned ICE-based ATV into a safe and efficient electric all-terrain vehicle.

This involves replacing the internal combustion engine with an electric powertrain, strengthening the chassis, and upgrading the steering, suspension, braking, and electrical systems.

The project also aims to reduce emissions, noise, and operating costs while effectively reusing the existing structure for off-road applications and academic demonstration.

2. LITERATURE REVIEW

2.1 Review of Existing ATV Designs

All-Terrain Vehicles (ATVs) are specially designed vehicles capable of operating on rough and uneven terrains such as mud, sand, rocks, and forest trails. Traditional ATV designs are primarily based on internal combustion engine (ICE) systems, which provide high power and torque required for off-road conditions. The chassis of an ATV is generally a tubular space frame structure that offers strength, rigidity, and lightweight characteristics.

Figure 2.1 ATV Design FV

Figure 2.1.1 ATV Design SV

The suspension system, commonly double wishbone type, is designed to absorb shocks and maintain stability over uneven surfaces. Steering systems such as Ackermann geometry are widely used to ensure proper turning radius and maneuverability. However, conventional ATV designs focus more on performance rather than sustainability, resulting in increased fuel consumption and environmental impact.

In recent years, researchers have started exploring electric ATV designs, which integrate electric motors, battery packs, and controllers while maintaining the structural integrity and off-road capability of traditional ATVs.

2.2 Research Papers Review

Various research studies have contributed significantly to the development of electric ATVs. Recent papers focus on the design and development of high-performance electric ATVs with efficient powertrains and optimized chassis structures. These studies highlight the importance of selecting suitable electric motors, battery capacity, and control systems to achieve desired performance.

Other research emphasizes the complete vehicle design, including suspension, steering, and braking systems, ensuring safety and durability in off-road conditions. Powertrain-focused studies discuss tractive effort calculations, torque requirements, and energy efficiency.
Additionally, structural analysis papers utilize Finite Element Analysis (FEA) to optimize the ATV frame and reduce weight while maintaining strength. Research on battery systems explores lithium-ion battery configurations, thermal management, and vibration resistance for off-road applications. Steering system studies focus on improving maneuverability using rack and pinion mechanisms with Ackermann geometry.

Overall, these research papers provide a strong foundation for designing a reliable and efficient electric ATV.

2.3 Existing ICE ATV Systems

Conventional ATVs are powered by internal combustion engines that operate on petrol or diesel. These systems consist of multiple components such as engine, fuel tank, carburetor or fuel injection system, exhaust system, and transmission. ICE-based ATVs are known for their high power output and long range; however, they suffer from several drawbacks.

They produce harmful emissions such as carbon monoxide (CO), hydrocarbons (HC), and nitrogen oxides (NOx), contributing to environmental pollution. Additionally, these vehicles generate significant noise and require frequent maintenance due to complex mechanical components.

The efficiency of ICE engines is relatively low compared to electric systems, and the dependency on fossil fuels increases operating costs. Due to these limitations,

there is a growing need to replace ICE systems with cleaner and more efficient alternatives.

2.4 EV Conversion Concepts

Electric Vehicle (EV) conversion involves replacing the internal combustion engine and associated components with an electric powertrain. In the case of an ATV, this includes installing an electric motor, battery pack, motor controller, and necessary electrical systems.
The electric motor provides instant torque, which is highly beneficial for off-road applications. Battery packs, typically lithium-ion, are designed to provide sufficient energy for desired range and performance. The motor controller regulates power flow between the battery and motor, ensuring smooth operation.

EV conversion also involves modifying the chassis to accommodate battery placement and ensuring proper weight distribution. Additionally, electrical safety, wiring insulation, and thermal management are important considerations.

This approach not only reduces emissions and noise but also lowers operating costs and promotes sustainable transportation.

2.5 Summary of Literature

From the literature review, it is observed that electric ATVs offer a promising alternative to conventional ICE-based vehicles. The integration of electric powertrains improves efficiency, reduces emissions, and minimizes maintenance requirements.

Research studies emphasize the importance of proper design of chassis, suspension, steering, and battery systems to achieve optimal performance. Structural analysis and optimization techniques ensure safety and durability under off-road conditions.

Furthermore, EV conversion of existing ATV structures is a cost-effective and environmentally friendly approach, as it utilizes existing resources while upgrading the vehicle to modern standards.

Hence, the present project focuses on developing an electric ATV by incorporating an efficient electric drive system, improved structural design, and enhanced safety features based on insights gained from the literature.

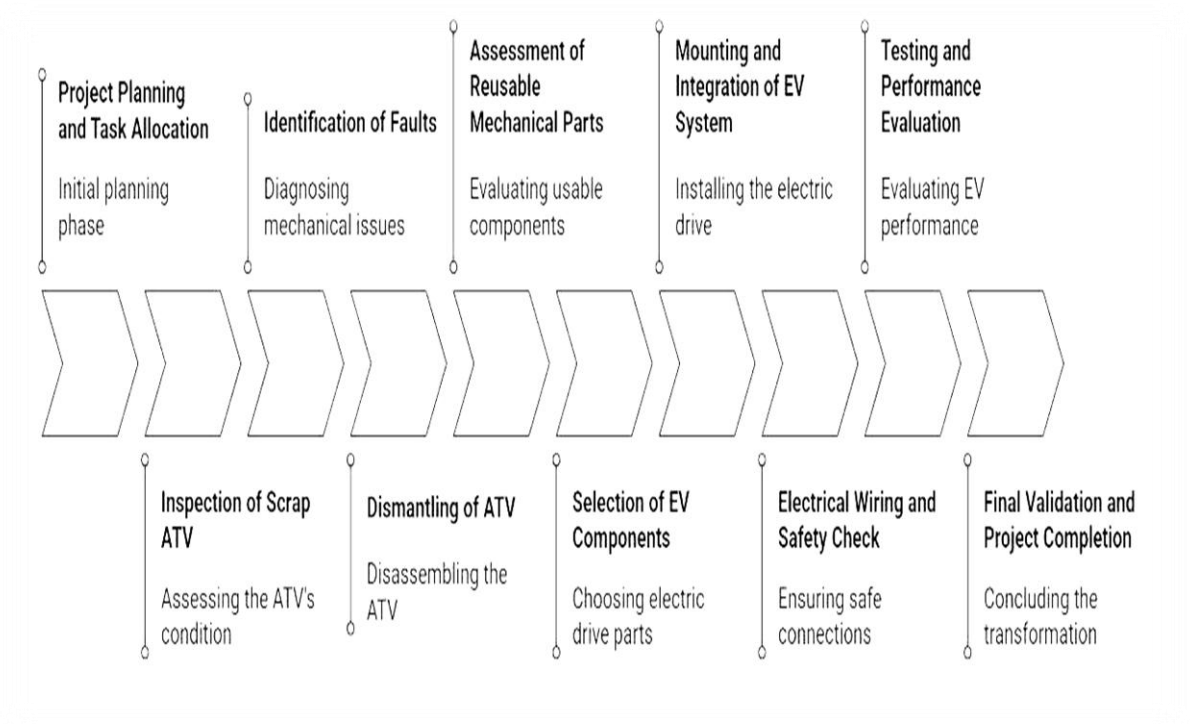
3. METHODOLOGY

3.1 Project Planning and Flow Chart

Project planning is an essential step in the successful development of the Electric All-Terrain Vehicle (ATV). It involves systematic scheduling of activities, proper utilization of resources, and coordination among project members. The planning ensures that the conversion of an old IC engine scrap ATV into an Electric ATV is completed within the given time, cost, and technical constraints.

The project was planned in different phases starting from inspection of the existing ATV, removal of IC engine components, selection of electric vehicle components, system integration, and final testing. Each phase was carefully executed to avoid errors and ensure safety and performance of the electric ATV.

Proper planning helped in minimizing material wastage, reducing fabrication errors, and achieving the desired project objectives.



3.2 Component Selection

Component selection plays a vital role in achieving the required performance, efficiency, and reliability of the Electric All-Terrain Vehicle (ATV). The components are selected after considering factors such as availability, cost, efficiency, durability, load capacity, and compatibility with the existing ATV structure.

Major Components Selection for Electric ATV

Battery (Lithium-ion / Lead-acid):
The battery is selected to provide adequate electrical energy required for the operation of the vehicle under varying load and terrain conditions. Lithium-ion batteries are preferred due to their high energy density, lightweight nature, and longer life cycle, whereas lead-acid batteries offer a cost-effective alternative.

Electric Motor (BLDC / DC Motor):
The electric motor is chosen based on the required torque and speed characteristics suitable for off-road applications. Brushless DC (BLDC) motors are generally preferred due to their high efficiency, low maintenance, and better performance compared to conventional DC motors.

Motor Controller:
The motor controller is used to regulate the speed and torque of the motor based on throttle input. It ensures smooth acceleration, efficient power delivery, and protection of the electrical components.

Chassis Material (Mild Steel):
Mild steel is selected as the chassis material due to its high strength, good weldability, durability, and ease of availability. It also provides the necessary structural rigidity required for off-road conditions.

Suspension System:
The suspension system is designed to absorb shocks and vibrations encountered during rough terrain operation. It improves vehicle stability, ride comfort, and handling performance.
Braking System (Disc Brake):

Disc brakes are selected to provide efficient and reliable braking performance under varying load and terrain conditions. They offer better heat dissipation and braking efficiency compared to drum brakes.

Steering Mechanism:
The steering mechanism is designed to ensure smooth handling, precise control, and improved maneuverability of the ATV, especially during off-road driving conditions.

3.4 Fabrication Method

The fabrication of the Electric All-Terrain Vehicle (E-ATV) involves a systematic process of modifying the existing chassis and integrating the required electric components. The aim is to ensure structural strength, proper alignment, and efficient functioning of all systems.

Initially, the existing ATV chassis is inspected, cleaned, and repaired to remove rust and damaged sections. Necessary modifications are made to accommodate the electric powertrain components such as the motor and battery pack.

The fabrication process includes cutting, welding, drilling, and finishing operations. Mounting brackets are fabricated to securely install the motor, battery, and controller. Proper alignment of drivetrain components is ensured to avoid power losses and mechanical failures.

Electrical wiring is carried out carefully with proper insulation and routing to ensure safety and reliability. All components are assembled and tested for fitment and functionality.

Finally, surface finishing such as painting or coating is applied to protect the chassis from corrosion and enhance its appearance. The complete fabrication process ensures that the vehicle is robust, safe, and suitable for off-road applications.

3.5 Testing Method

Testing is performed to evaluate the performance, safety, and reliability of the Electric ATV. Both static and dynamic tests are conducted.

Testing Includes:

- Battery voltage and current test
- Motor performance test
- Brake efficiency test
- Steering and suspension test
- Load and speed test

Testing ensures that the vehicle operates smoothly under different conditions and meets design objectives.

3.6 Safety Considerations

Safety is given top priority during the design and development of the Electric ATV. Protective measures are implemented for both electrical and mechanical systems.

Safety Measures:

- Proper insulation of electrical wiring
- Fuse and emergency cutoff switch
- Secure mounting of battery and motor
- Protective covers for rotating parts
- Reliable braking system

These safety considerations help prevent accidents and ensure safe operation of the Electric ATV.

4. SYSTEM DESCRIPTION

4.1 Overall Block Diagram

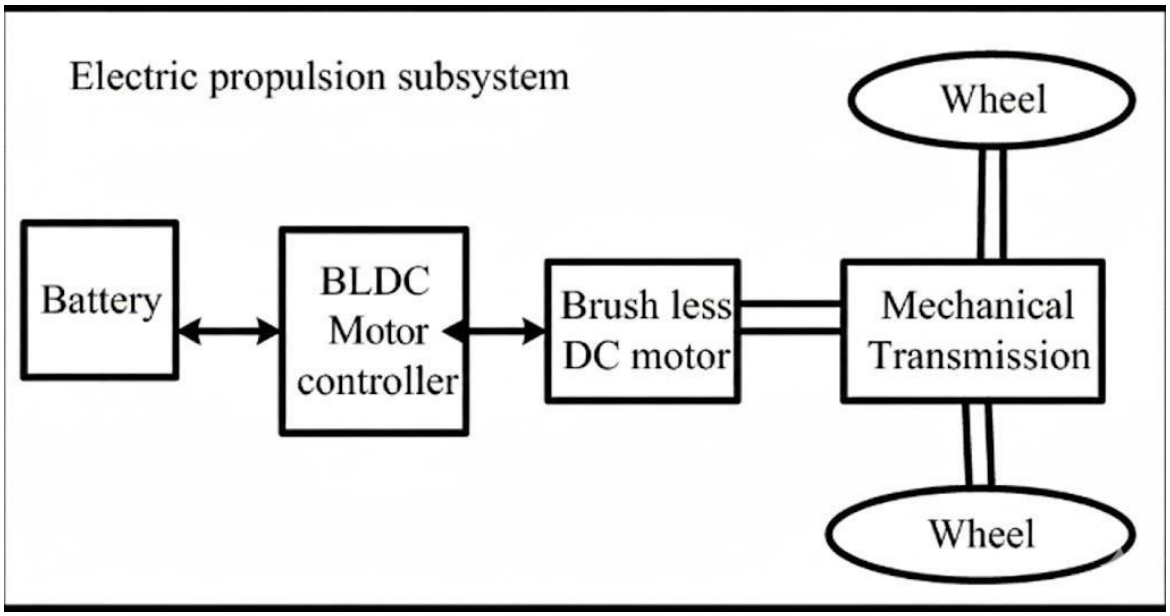


Figure 4.1 Block Diagram

4.2 Electric Propulsion System of EV ATV

The electric propulsion system of the Electric All-Terrain Vehicle (E-ATV) consists of key components such as the battery pack, BLDC motor controller, BLDC motor, mechanical transmission system, and wheels. These components work together to convert electrical energy into mechanical motion for vehicle propulsion.

The battery pack acts as the primary energy source and supplies direct current (DC) electrical power to the motor controller. The motor controller plays a crucial role in regulating the voltage and current supplied to the BLDC motor based on the throttle input provided by the driver.

The BLDC motor then converts the electrical energy into mechanical rotational energy with high efficiency and reliability. This rotational motion is transmitted to the wheels through a mechanical transmission system, such as a Continuously Variable Transmission (CVT), which helps in maintaining optimal torque and speed under different operating conditions.

Finally, the transmitted mechanical power drives the wheels, resulting in smooth and controlled movement of the vehicle across various terrains.

4.3 Steering System

The steering system of the All-Terrain Vehicle (ATV) is used to control the direction of the vehicle by turning the front wheels. In this project, a rack and pinion steering mechanism is employed due to its simple construction, compact size, and smooth operation.

When the driver rotates the steering wheel, the motion is transmitted through the steering shaft to the pinion gear. The pinion gear moves the rack linearly, which turns the front wheels left or right.

Proper steering geometry ensures vehicle stability, manoeuvrability, and safe handling during off-road conditions.

Figure 4.3 Steering system

4.4 Braking System

The braking system of the All-Terrain Vehicle (ATV) is used to reduce speed and stop the vehicle safely. In this project, a hydraulic disc braking system is employed due to its high braking efficiency and reliable performance.

When the brake pedal is pressed, hydraulic fluid transmits pressure to the brake calliper. The calliper forces the brake pads against the rotating disc, creating friction that slows down the wheel.

This system provides effective braking even under off-road and uneven surface conditions.

Hydraulic disc brakes provide quick response and better heat dissipation compared to drum brakes.

The system requires less maintenance and offers consistent braking performance. Disc brakes are effective in wet, muddy, and dusty conditions, making them suitable for ATV applications.

Figure 4.4 Breaking system

4.5 Suspension System

The suspension system used in the ATV consists of a coil spring with shock absorber assembly and independent front suspension arms.

The suspension absorbs shocks and vibrations produced due to uneven and off-road surfaces.

The coil spring supports the vehicle load, while the shock absorber controls spring oscillations and prevents excessive bouncing.

The suspension arms connect the wheel hub to the chassis, allowing controlled vertical movement of the wheels. This system improves ride comfort, maintains tyre contact with the ground, and enhances vehicle stability during operation.

The coil spring supports the vehicle load, while the shock absorber controls spring oscillations and prevents excessive bouncing.

The suspension arms connect the wheel hub to the chassis, allowing controlled vertical movement of the wheels.

This suspension system improves ride comfort, maintains continuous tyre contact with the ground, and enhances overall vehicle stability during operation.

Figure 4.5 Suspension System

4.6 Battery Pack

The battery pack acts as the energy storage unit of the Electric ATV. It supplies electrical power to the motor through the motor controller. Lithium-ion batteries are preferred due to their high energy density, light weight, and longer life compared to conventional batteries.

The battery pack is designed to deliver the required voltage and current for the desired vehicle range and performance.

Proper casing, insulation, and secure mounting are provided to ensure electrical and mechanical safety.

The battery pack plays a vital role in determining the efficiency, reliability, and overall performance of the EV ATV.

Figure 4.6Battery pack

4.7 Electric Motor

The electric motor is the prime mover of the Electric All-Terrain Vehicle (EV ATV). It converts electrical energy supplied by the battery into mechanical energy to drive the vehicle. In this project, a BLDC (Brushless Direct Current) motor is used due to its high efficiency, compact size, and low maintenance.
The electric motor provides the required torque and speed for off-road operation of the EV ATV.

BLDC motors offer high efficiency, smooth operation, and longer service life as they do not use brushes.

The motor is coupled with a mechanical transmission system to transfer power effectively to the wheels.

The electric motor ensures smooth, quiet, and emission-free operation compared to conventional internal combustion engines.

Figure 4.7 Electric Motor

4.8 Motor Controller

The motor controller is an essential component of the Electric All-Terrain Vehicle (EV ATV) as it controls the operation of the electric motor. It acts as an interface between the battery pack and the electric motor. The motor controller receives electrical power from the battery and regulates the voltage and current supplied to the BLDC motor based on throttle input.

The motor controller controls the speed, torque, and direction of rotation of the motor. It also provides protection against over-current, over-voltage, and short-circuit conditions, ensuring safe and reliable operation of the EV ATV. The controller improves overall efficiency and smoothness of vehicle operation.

The motor controller regulates power flow from the battery to the BLDC motor.

It controls motor speed and torque according to accelerator input.

Built-in protection features enhance safety and increase component life.

Figure 4.8 Motor Controller

4.9 Throttle

The throttle is an input control device used to regulate the speed of the Electric All-Terrain Vehicle (EV ATV). It sends an electrical signal to the motor controller, which controls the power supplied to the BLDC motor. In this project, an electronic throttle with a hall-effect sensor is used for smooth operation.

It controls vehicle speed by varying motor power.
It provides smooth and gradual acceleration.
Electronic throttle requires less maintenance and improves safety.

Figure 4.9 Throttle

4.10 Charging System

The charging system of the Electric All-Terrain Vehicle (EV ATV) is used to recharge the battery pack from an external electrical power source. In this project, an on-board/off-board battery charger is used to convert AC power from the mains supply into DC power suitable for charging the battery.

The charger supplies controlled voltage and current to ensure safe charging of the battery pack.

Proper connectors and protection devices are used to avoid overcharging and electrical faults.

The charging system improves convenience and ensures reliable operation of the EV ATV.

Figure 4.10 Charging System

4.11 CVT

The Continuously Variable Transmission (CVT) is used in the EV ATV to transmit power from the electric motor to the wheels while providing smooth and variable speed ratios. Unlike conventional gearboxes, the CVT does not use fixed gears; instead, it uses a belt and pulley system to provide continuous variation in speed and torque.

The CVT automatically adjusts the gear ratio based on load and speed conditions. It provides smooth acceleration and better torque delivery, which is suitable for offroad operation.
The CVT improves driving comfort and reduces the need for manual gear shifting.

Figure 4.11 CVT

4.12 Hydraulic Brake

The hydraulic brake system is used in the EV ATV to safely reduce speed and stop the vehicle. It operates on Pascal's law, where brake force is transmitted through hydraulic fluid. When the brake pedal is pressed, hydraulic pressure is generated in the master cylinder and transmitted to the brake calliper. The calliper presses the brake pads against the disc, creating friction that slows or stops the wheel.

Hydraulic brakes provide strong and uniform braking force.

They offer quick response and better control, especially in off-road conditions.

Disc brakes work effectively in muddy, wet, and dusty environments.

Figure 4.12 Braking system

4.13 Chassis

The chassis is the main structural frame of the Electric All-Terrain Vehicle (EV ATV) on which all components are mounted. It supports the weight of the battery pack, motor, transmission, steering, braking, and suspension systems. In this project, a scrap ATV chassis is reused, as it provides sufficient strength and durability for off-road operation.

The chassis provides structural strength and rigidity to the vehicle.

It maintains proper alignment of all systems and ensures vehicle stability.

Reusing a scrap chassis reduces cost and material wastage, making the project economical.

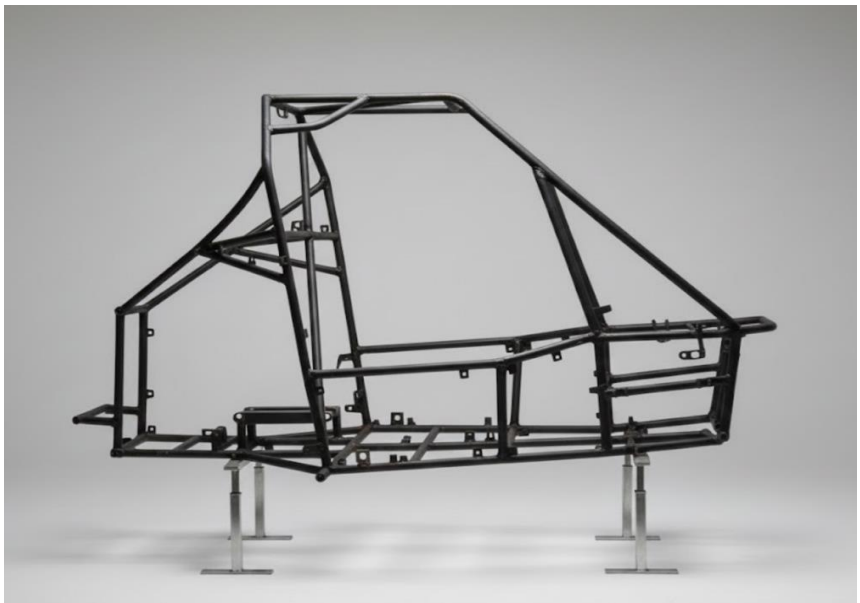


Figure 4.13Chassis

5. DESIGN CALCULATIONS

5.1 Vehicle Speed Calculation

The theoretical speed of the electric ATV is calculated using:

$$V = \frac{\pi D N}{60 \times G}$$

Where:

D= Wheel diameter (m)

N= Motor speed (RPM)

G= Gear ratio

This equation helps determine the final vehicle speed after gear reduction. Electric ATVs typically operate in the range of 30–50 km/h for safe off-road performance.

5.2 Torque Calculation

Torque is critical for off-road capability. It is calculated as:

$$T = \frac{P \times 60}{2\pi N}$$

Where:

P= Power (W)

N= Motor speed (RPM)

High torque enables the ATV to climb slopes and traverse rough terrain effectively. Gear reduction is used to increase torque at the wheels.

5.3 Power Requirement

The total power required is given by:

$$P = (F_r + F_g + F_a) \times V$$

Where:

F_r= Rolling resistance

F_g= Gradient resistance

F_a= Aerodynamic drag

For an electric ATV, a BLDC motor of 1000W–2000W is generally sufficient.

5.4 Battery Capacity Calculation

Battery capacity is calculated as:

$$Ah = \frac{P \times t}{V}$$

Where:
P= Power (W)
t= Time (hours)
V= Battery voltage
Typical configurations:
48V or 60V battery
30–50 Ah capacity

5.5 Range Estimation

Range is estimated using:

$$Range = \frac{Battery\ Energy}{Power\ Consumption}$$

In practical conditions, the electric ATV achieves a range of 40–70 km per charge, depending on terrain and load.

5.6 Braking Force Calculation

$$F_b = \mu \times m \times g$$

Where:
μ= Coefficient of friction
m= Mass of vehicle
g= Acceleration due to gravity
Disc brakes are preferred for better stopping performance and safety.

5.7 Shaft Design

The shaft diameter is determined using:

$$T = \frac{\pi}{16}\tau d^3$$

Where:

τ= Shear stress

d= Diameter of shaft

Material selection (e.g., Mild Steel or EN8) and a proper factor of safety are essential for durability.

6. FABRICATION PROCESS

6.1 Tools and Equipment Used

Fabrication of the EV ATV requires both hand tools and heavy machinery to ensure accurate construction, safety, and durability. Key tools include:

Spanners and Wrenches:
Used for tightening nuts and bolts during assembly.

Angle Grinder:
For cutting, grinding, and smoothing mild steel pipes or sheets.

Welding Machine:
Used for joining frame components, brackets, and other metal parts securely. MIG or arc welding can be used depending on material thickness.

Drilling Machine:
For creating precise holes for bolts, mounting brackets, and components.

Lathe Machine:
Used to machine shafts, sprockets, and other rotational components with accuracy.

Measuring Instruments:
Vernier callipers, micro meters, and measuring tapes ensure dimensional accuracy.

6.2 Frame Manufacturing

The ATV frame forms the backbone of the vehicle, supporting all components and the rider.

Material Used:
Mild steel (MS) pipes are preferred for their strength, weldability, and cost-effectiveness.

Fabrication Process:
Cutting: Pipes are cut to required lengths using an angle grinder or cutting machine.

Bending:
Pipes are bent to achieve the required geometry for the ATV frame. A pipe-bending machine or jig is used for accuracy.

Welding:
The cut and bent pipes are joined using MIG/arc welding. All weld joints are checked for strength and defects.

Reinforcements:
Critical load-bearing areas, like suspension mounts and motor brackets, are reinforced with gussets.
Surface Finishing:
The frame is ground, cleaned, and coated with primer or paint to prevent corrosion.

Design Considerations:
The frame must withstand off-road loads, vibrations, and shocks.
Weight optimization is critical for better acceleration and range of the EV ATV.

6.3 Mounting of Motor and Battery
Proper mounting ensures stability, safety, and performance:

Motor Mounting:
The BLDC motor is mounted near the rear axle for direct torque transmission.
Steel brackets and bolts secure the motor to the frame.
Alignment with the sprocket/chain is critical to avoid excessive wear.

Battery Placement:
The battery is mounted at the centre of gravity to maintain stability.
Protective casing is used to prevent damage from vibration, shocks, or water ingress.
Proper wiring and connectors ensure safety and ease of maintenance.

Safety Considerations:
All mounting points are reinforced.
Vibration dampers may be used to protect electrical components.

6.4 Assembly of Steering

The steering system allows smooth maneuverability and safe handling:

Components:
Steering wheel
Steering shaft
Tie rods
Kingpins and wheel hubs

Assembly Process:
Steering shaft is aligned with front wheels.
Tie rods connect the steering shaft to wheel hubs.
Ackermann Steering Geometry ensures proper turning radius and reduces tire wear.

Design Considerations:
The steering column should be rigid yet slightly flexible to absorb shocks.
Bearings and lubrication reduce friction for smooth operation.

6.5 Brake Installation

Brakes are crucial for safety and control of the EV ATV:

Components:
Disc rotors on rear wheels
Hydraulic master cylinder
Brake pads
Brake lines and reservoirs
Installation Process:
Disc rotors are mounted on rear axle hubs.
Hydraulic lines are connected to master cylinder and brake calipers.
Brake fluid is filled, and the system is bled to remove air.

Key Considerations:
Brake torque should be sufficient for rapid deceleration without wheel lock-up.
Regular inspection is necessary for off-road conditions.

6.6 Electrical Wiring

The electrical system integrates all power and control components of the EV ATV:

Components:
Battery pack (48 V)
BLDC motor
Motor controller
Throttle sensor
Fuses and wiring harnesses

Wiring Process:
Battery Connection:
Positive and negative terminals are connected through a main fuse.
Motor Controller Wiring:
Motor phases and hall sensors connected to controller.

Throttle Wiring:
Signal wires from the throttle to the controller are routed safely.
Insulation: All wires are insulated and routed to avoid heat and friction.
Safety Measures: Fuses prevent overcurrent; proper earthing avoids short circuits.

Key Considerations:
Avoid sharp bends and rubbing against metal edges.
Maintain accessibility for maintenance and troubleshooting.

6.7 Final Assembly

Steps:
Frame, motor, battery, suspension, wheels, brakes, and steering are fully assembled.
Torque all bolts and nuts to recommended specifications.
Electrical system is tested for continuity, voltage, and proper operation.
Mechanical components are checked for free rotation and alignment.

The vehicle is tested under static and dynamic conditions before road testing.

Final Checks:
Suspension travel and alignment
Brake efficiency
Steering response
Electrical safety (no loose wires, proper insulation)

Outcome:
The EV ATV is ready for testing with all components integrated, ensuring mechanical reliability, electrical safety, and optimal performance.

7. TESTING & PERFORMANCE ANALYSIS

Testing is a critical phase in the development of the Electric ATV as it validates the design, ensures safety, and evaluates performance under real-world conditions. Both mechanical and electrical systems are tested for reliability, efficiency, and durability to ensure that the vehicle meets its design specifications.

7.1 No load Test

The no-load test is the first step in the evaluation process. In this test, the ATV is operated without any external load to assess the motor and electrical system performance. The motor is gradually operated at different throttle levels, and key parameters such as RPM, current, and voltage are recorded. This test confirms that the motor and controller are functioning correctly, and any unusual vibrations, noise, or heating are identified. The measured no-load RPM is compared with theoretical calculations based on the motor speed constant and battery voltage to verify motor efficiency.

7.2 load Test

Following the no-load test, the load test is conducted to evaluate the ATV's performance under actual operating conditions. This involves adding the rider and expected payload, then driving the vehicle on flat terrain and inclines. During this test, acceleration, speed, motor current, and battery voltage are recorded. The observed acceleration is compared with theoretical calculations based on traction force and resistive forces. This test ensures that the motor and battery can handle real-world loads without overloading the system or causing excessive voltage drops.

7.3 Speed Test

The speed test determines the actual top speed of the ATV and verifies the effectiveness of the gear ratios and motor power. The ATV is tested on a flat, straight track, and speed is measured using a speedometer or GPS device. Multiple runs are performed to ensure consistent results. The actual speed is then compared with the theoretical speed derived from the motor RPM and gear ratio, allowing assessment of efficiency losses due to friction, drag, or drivetrain limitations. This step also evaluates vehicle control and stability at higher speeds.

7.4 Brack Test

The brake test ensures that the ATV can stop safely under normal and emergency conditions. The braking system is evaluated by measuring stopping distance at different speeds, monitoring brake torque, and checking for smooth operation. Both the hydraulic disc brakes and regenerative braking system of the motor are tested. The stopping distance is compared to theoretical calculations based on vehicle mass and braking force. The hydraulic brakes demonstrated smooth and controlled deceleration without wheel lock-up, while regenerative braking assisted in slowing the vehicle and recovering energy.

7.5 Stability Test

Stability tests are conducted to ensure the ATV can safely navigate turns, slopes, and uneven terrain without tipping. The ATV is driven on off-road paths, and the response of the suspension, weight distribution, and center of gravity are observed. Maximum tilt angles during turns are evaluated to check rollover resistance. Proper placement of the battery and motor lowers the center of gravity, which significantly improves stability. The suspension system effectively absorbs shocks, maintaining control and comfort over rough terrain.

7.6 Results and discussion

The results and discussion from all tests indicate that the EV ATV met design expectations. The motor operated within safe limits, achieving the desired speed and acceleration, while the braking system ensured safety with controlled stopping distances. Stability was maintained during off-road testing, and the suspension efficiently absorbed shocks. The electrical system performed reliably, with minimal voltage drops and no issues in wiring or insulation. Overall, the vehicle demonstrates safe, reliable, and efficient operation, suitable for off-road conditions, while minor improvements in suspension damping and regenerative braking could further enhance performance.

8. COST ESTIMATION

8.2Electrical Components Cost

Material Name	Quantity	Prize
Battery	01	₹ 31000
Electric Motor	01	₹18000
Motor Controller	01	Free with motor

₹ 49000

9. ADVANTAGES, LIMITATIONS & APPLICATIONS

9.1 Advantages of ATV

The All-Terrain Vehicle (ATV), especially in its electric configuration, offers several advantages over conventional vehicles in off-road and utility applications. One of the primary advantages is its high maneuverability. Due to its compact size, low turning radius, and lightweight design, the ATV can easily navigate through rough terrains such as mud, sand, rocky paths, and forest trails where conventional vehicles may struggle. This makes it highly suitable for off-road operations.

Another significant advantage is its eco-friendly operation in the case of an electric ATV. Since it uses a battery-powered system instead of an internal combustion engine, it produces zero tailpipe emissions, reduces noise pollution, and contributes to environmental sustainability. The ATV also provides better traction and stability on uneven surfaces. With wide tires and a low centre of gravity, the vehicle maintains good grip and balance even on slopes and irregular terrain, enhancing safety during operation.

Additionally, the ATV has low operating and maintenance costs. Electric motors have fewer moving parts compared to IC engines, resulting in reduced wear and tear, minimal servicing requirements, and lower running costs. The design of the ATV allows for versatility and adaptability. It can be modified or equipped with attachments for various tasks such as cargo transport, spraying, towing, and agricultural work.

9.2 Limitations

Despite its advantages, the ATV also has certain limitations that must be considered. One of the main limitations is its limited range in electric variants. Since the vehicle depends on battery capacity, it can only operate for a specific duration before requiring recharging, which may not be suitable for long-distance operations.

Another limitation is the lower payload capacity compared to larger vehicles. ATVs are designed to be lightweight and compact, which restricts the amount of load they can carry. The ATV may also face stability issues at high speeds, especially on uneven or inclined terrain. Due to its narrow structure, there is a risk of tipping if not handled properly.

Charging infrastructure can be another challenge, leading to longer downtime. Unlike fuel-powered vehicles that can be refuelled quickly, electric ATVs require significant time to recharge. Furthermore, the vehicle is less suitable for highway or urban transportation, as it is primarily designed for off-road conditions and may not comply with road safety regulations in some regions.

9.3 Applications

ATVs are widely used across multiple sectors due to their versatility and off-road capability. In the agricultural sector, ATVs are used for tasks such as transporting tools, spraying pesticides, monitoring crops, and carrying small loads across fields with uneven terrain.

In military and defence operations, ATVs are utilized for patrolling, reconnaissance, and transportation in remote and difficult terrains where larger vehicles cannot operate effectively. They are also commonly used in forestry and wildlife management for surveillance, maintenance of forest areas, and rescue operations.

In the field of recreation and sports, ATVs are popular for off-road racing, adventure activities, and trail riding, providing an exciting driving experience. Additionally, ATVs are used in industrial and construction sites for material handling and movement within rough and inaccessible areas. Electric ATVs are increasingly being adopted in eco-sensitive zones, resorts, and campuses due to their silent operation and zero emissions.

10. FUTURE SCOPE

The developed Electric All-Terrain Vehicle (EV ATV) demonstrates significant potential for further improvements and advancements. With rapid developments in electric mobility and automotive technologies, the performance, safety, and efficiency of the ATV can be enhanced in several ways. The following sections highlight the future scope of the project.

10.1 Design Improvements

The current ATV design can be further optimized to improve performance, durability, and user comfort. One of the major areas of improvement is weight reduction, which can be achieved by using advanced materials such as aluminium alloys, composites, or high-strength lightweight steel. Reducing the vehicle weight will directly improve efficiency, speed, and battery life.

The aerodynamic design can also be enhanced, although ATVs are primarily off-road vehicles. Streamlining the body structure can reduce drag and improve overall energy efficiency during motion. Ergonomic improvements such as better seating position, handlebar design, and control placement can enhance driver comfort and reduce fatigue during long-duration use.

Additionally, the chassis design can be further strengthened using advanced simulation tools like Finite Element Analysis (FEA) to ensure better structural integrity and durability under extreme conditions.

10.2 Battery Upgrade

Battery technology plays a crucial role in the performance of an electric ATV. Future improvements can include the use of higher capacity lithium-ion batteries to increase the driving range and reduce the frequency of charging. The implementation of a Battery Management System (BMS) can significantly improve battery performance, safety, and lifespan by monitoring parameters such as voltage, temperature, and state of charge.

Fast-charging technology can be incorporated to reduce charging time, making the vehicle more practical for real-world applications. In the future, emerging technologies such as solid-state batteries may be adopted, offering higher energy density, improved safety, and longer life cycles compared to conventional batteries.

10.3 Suspension Improvement

The suspension system of the ATV can be enhanced to provide better ride comfort and stability on rough terrains. Advanced suspension systems such as independent suspension for all wheels can improve shock absorption and maintain better tire contact with the ground. Adjustable suspension systems can be introduced to allow the rider to modify ride height and stiffness based on terrain conditions such as mud, sand, or rocky surfaces.

The use of high-performance shock absorbers and dampers can reduce vibrations and improve overall handling and control of the vehicle. Further improvements can include the integration of adaptive or semi-active suspension systems, which automatically adjust damping characteristics based on terrain feedback.

10.4 Advanced Safety Features

Safety can be significantly improved by integrating modern technologies into the ATV design. Features such as disc brakes with improved callipers and better braking materials can enhance stopping performance. The inclusion of electronic safety systems like traction control can help maintain stability on slippery or uneven surfaces by preventing wheel slip.

A roll-over protection structure (ROPS) can be implemented to protect the rider in case of vehicle overturning, which is a common risk in off-road vehicles. Additional features such as LED lighting systems, reverse alarms, and digital displays can improve visibility and driver awareness.

In future developments, smart safety systems like GPS tracking, remote monitoring, and emergency alert systems can also be incorporated to enhance operational safety and vehicle tracking.

11. CONCLUSION

The design and development of the Electric All-Terrain Vehicle (ELC ATV) has been successfully completed by integrating both mechanical and electrical engineering principles. The project demonstrates the feasibility of using electric propulsion systems for off-road applications.

The ATV was designed with a strong and stable frame capable of withstanding rough terrain conditions. The selection of a BLDC motor provided high torque at low speeds, making the vehicle suitable for climbing slopes and navigating uneven surfaces. The battery system was carefully chosen to balance power, range, and weight, ensuring efficient performance.

Various design calculations such as speed, torque, power requirement, and battery capacity were carried out to ensure optimal performance of the vehicle. The fabrication process was performed using standard manufacturing techniques, and all components were assembled with proper alignment and safety considerations.

Testing of the ATV under different conditions, including no-load, load, speed, and braking tests, showed satisfactory performance. The vehicle demonstrated good stability, adequate speed, and efficient energy utilization. The braking system provided reliable stopping performance, and the overall structure remained stable during operation.

This project highlights the advantages of electric vehicles, including eco-friendliness, low operating cost, and reduced maintenance. Although certain limitations such as limited range and charging time exist, the results confirm that electric ATVs are a promising solution for future transportation, especially in off-road and rural applications.

In conclusion, the developed electric ATV is a cost-effective, efficient, and sustainable alternative to conventional fuel-powered vehicles, and it provides a strong foundation for further improvements and innovations in electric mobility.

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13. After development

Before

Figure 13.1 Before Development

After Development

Figure 13.2 After Development